

UNSW School of Mathematics and Statistics

MATH2601 Higher Linear Algebra

Tutorial problems, Week 2

1. Write down the multiplication table for the group of symmetries of the equilateral triangle.

(Careful – the group is not abelian, so set the table up so that an entry in a row is multiplied on the **left** of an entry in a column.)

2. Find all solutions to the equation $x^2 - 5x + 4 = 0$ in $\mathbb{Z}_{10}$. Why are there more than two solutions?
3. Given that there is a field with exactly four elements $\mathbb{F} = \{0, 1, \alpha, \beta\}$, where 0 is the additive identity and 1 is the multiplicative identity, find the addition and multiplication tables for $\mathbb{F}$. (Note: the addition table is the hard part!)
4. Recall from the previous week the group $G = \{M_a : a \in \mathbb{R}\}$ where

$$M_a = \begin{pmatrix} 1 & a & \frac{1}{2}a^2 \\ 0 & 1 & a \\ 0 & 0 & 1 \end{pmatrix}.$$

Let $\mathbb{R}$ be the group of real numbers under addition. Show that

$$\phi : \mathbb{R} \rightarrow G \quad \text{where} \quad \phi(a) = M_a$$

is a group homomorphism. Is it an isomorphism?

5. Let A be a fixed $n \times n$ real matrix. Show that

$$S = \{X \in M_{n,n}(\mathbb{R}) : AXA = X\}$$

is a vector space.

6. Let V be a vector space, let $\mathbf{v}, \mathbf{w}$ be vectors in V , and let α be a scalar in the associated scalar field. Prove, giving a reason for each step, that if $\alpha\mathbf{v} = \alpha\mathbf{w}$ and $\alpha \neq 0$, then $\mathbf{v} = \mathbf{w}$.